DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Master of Technology (Computer Science & Information Technology) SEMESTER - 3 Winter 2025 (Regular)

Course :Master of Technology (Computer Science & Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: MTCSEMD302F - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:
1.All the questions are compulsory.
2.Use of non-programmable scientific calculators is allowed.
3.Assume suitable data wherever necessary and mention it clearly.

1	Solve Any Two of the following.		
	Q1(a)	Compare BFS, DFS, and A* search in terms of time, space, completeness, and optimality.?	6
	Q1(b)	Discuss the challenges of applying classical search techniques to large real-world AI problems.?	6
	Q1(c)	Differentiate between unguided and guided search techniques with examples?	6
2	Solve Any Two of the following.		
	Q2(a)	Explain the structure of a neural network and the learning process in BPNN?	6
	Q2(b)	Describe the working of a Genetic Algorithm with a suitable optimization problem?	6
	Q2(c)	Critically analyze gradient descent learning versus evolutionary learning approaches?	6
3	Solve Any Two of the following.		
	Q3(a)	Explain AI planning and its components?	6
	Q3(b)	Explain game playing as a search problem with evaluation functions?	6
	Q3(c)	Is admissibility always necessary in heuristic search? Discuss with justification.?	6
4	Solve Any Two of the following.		
	Q4(a)	Explain predicate logic and its role in knowledge representation?	6
	Q4(b)	How can iterative deepening be applied to minimax search procedure?	6
	Q4(c)	Discuss limitations of Minimax and the need for advanced game-playing techniques?	6
5	Solve Any Two of the following.		
	Q5(a)	Explain fuzzy logic and certainty factors?	6
	Q5(b)	Describe expert system architecture and development steps?	6
	Q5(c)	Discuss integration of machine learning in expert systems and its challenges?	6

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